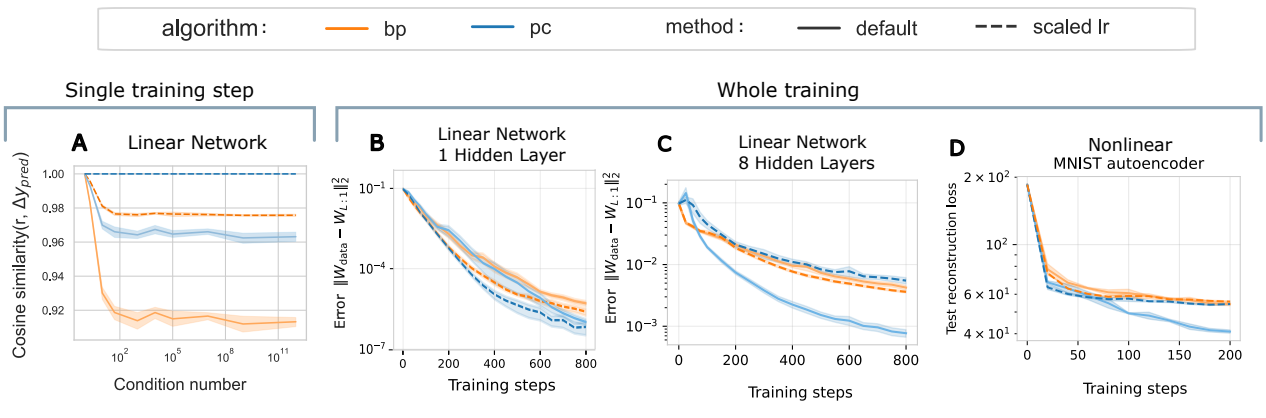

algorithm:
bp
pc
method:
default
scaled lr
Single training step
Whole training
A
Linear Network
Cosine similarity(r, Δy_{pred})
1.00
0.98
0.96
0.94
0.92
10^2
10^5
10^8
10^{11}
Condition number
B
Linear Network
1 Hidden Layer
Error $\|W_{data} - W_{L:1}\|_2^2$
10^{-1}
10^{-4}
10^{-7}
0
200
400
600
800
Training steps
C
Linear Network
8 Hidden Layers
Error $\|W_{data} - W_{L:1}\|_2^2$
10^{-1}
10^{-2}
10^{-3}
0
200
400
600
800
Training steps
D
Nonlinear
MNIST autoencoder
Test reconstruction loss
2×10^2
10^2
6×10^1
4×10^1
0
50
100
150
200
Training steps